

Exercise 1D Solutions

Order of Operations and BODMAS Rules for Integers

The BODMAS Rule for Simplifying Expressions

To simplify mathematical expressions involving multiple operations, we strictly follow the order of operations given by the acronym **BODMAS**:

1. **B – Brackets**: Evaluated in the following priority:
 - (a) **Bar or Vinculum** ($\overline{x - y}$) – Operations under the bar are performed first.
 - (b) **Round Brackets or Parentheses** ()
 - (c) **Curly Brackets or Braces** { }
 - (d) **Square Brackets or Big Brackets** []
2. **O – Of**: Represents multiplication (e.g., a of $b = a \times b$).
3. **D – Division** ($\div$)
4. **M – Multiplication** ($\times$)
5. **A – Addition** ($+$)
6. **S – Subtraction** ($-$)

Complete Step-by-Step Solutions

Here are the detailed, step-by-step simplifications for each problem in **Exercise 1D**:

1. Simplify: $5\{(-6) + (12 \div 4)\}$

1. **Step 1: Simplify division inside the round brackets.**

$$12 \div 4 = 3$$

Substituting this back into the expression:

$$5\{(-6) + 3\}$$

2. **Step 2: Simplify addition inside the curly brackets.**

$$(-6) + 3 = -3$$

Substituting this back:

$$5\{-3\}$$

3. **Step 3: Perform multiplication.**

$$5 \times (-3) = -15$$

Answer: -15

2. Simplify: $10 - \{56 \div (-16 + 9)\}$

1. **Step 1: Simplify addition inside the round brackets.**

$$-16 + 9 = -7$$

Substituting this back into the expression:

$$10 - \{56 \div (-7)\}$$

2. **Step 2: Simplify division inside the curly brackets.**

$$56 \div (-7) = -8$$

Substituting this back:

$$10 - \{-8\}$$

3. **Step 3: Perform subtraction (double negative becomes addition).**

$$10 - (-8) = 10 + 8 = 18$$

Answer: 18

3. Simplify: $(-6) - \{(-28) \div (-7)\}$

1. **Step 1: Simplify division inside the curly brackets.** Since dividing two negative integers yields a positive quotient:

$$(-28) \div (-7) = 4$$

Substituting this back into the expression:

$$(-6) - \{4\}$$

2. **Step 2: Perform subtraction.**

$$-6 - 4 = -10$$

Answer: -10

4. Simplify: $8 - \{12 - (9 - \overline{7 - 4})\}$

1. **Step 1: Simplify operation under the bar/vinculum ($\overline{7 - 4}$).**

$$\overline{7 - 4} = 3$$

Substituting this back:

$$8 - \{12 - (9 - 3)\}$$

2. **Step 2: Simplify inside the round brackets.**

$$9 - 3 = 6$$

Substituting this back:

$$8 - \{12 - 6\}$$

3. **Step 3: Simplify inside the curly brackets.**

$$12 - 6 = 6$$

Substituting this back:

$$8 - 6 = 2$$

Answer: 2

5. Simplify: $11 - [(-8) - \{10 - (9 - \overline{7 - 4})\}]$

1. **Step 1: Simplify operation under the bar/vinculum ($\overline{7 - 4}$).**

$$\overline{7 - 4} = 3$$

Substituting this back:

$$11 - [(-8) - \{10 - (9 - 3)\}]$$

2. **Step 2: Simplify inside the round brackets.**

$$9 - 3 = 6$$

Substituting this back:

$$11 - [(-8) - \{10 - 6\}]$$

3. **Step 3: Simplify inside the curly brackets.**

$$10 - 6 = 4$$

Substituting this back:

$$11 - [(-8) - 4]$$

4. **Step 4: Simplify inside the square brackets.**

$$-8 - 4 = -12$$

Substituting this back:

$$11 - [-12]$$

5. **Step 5: Perform the final subtraction.**

$$11 - (-12) = 11 + 12 = 23$$

Answer: 23

6. Simplify: $(-9) - [(-5) - \{11 - (8 - \overline{5 - 2})\}]$

1. **Step 1: Simplify operation under the bar/vinculum ($\overline{5 - 2}$).**

$$\overline{5 - 2} = 3$$

Substituting this back:

$$(-9) - [(-5) - \{11 - (8 - 3)\}]$$

2. **Step 2: Simplify inside the round brackets.**

$$8 - 3 = 5$$

Substituting this back:

$$(-9) - [(-5) - \{11 - 5\}]$$

3. **Step 3: Simplify inside the curly brackets.**

$$11 - 5 = 6$$

Substituting this back:

$$(-9) - [(-5) - 6]$$

4. **Step 4: Simplify inside the square brackets.**

$$-5 - 6 = -11$$

Substituting this back:

$$(-9) - [-11]$$

5. **Step 5: Perform the final subtraction.**

$$-9 - (-11) = -9 + 11 = 2$$

Answer: 2

7. Simplify: $10 - [8 - \{11 + 30 \div (4 - \overline{5 - 7})\}]$

1. **Step 1: Simplify operation under the bar/vinculum ($\overline{5 - 7}$).**

$$\overline{5 - 7} = -2$$

Substituting this back:

$$10 - [8 - \{11 + 30 \div (4 - (-2))\}]$$

2. **Step 2: Simplify inside the round brackets.**

$$4 - (-2) = 4 + 2 = 6$$

Substituting this back:

$$10 - [8 - \{11 + 30 \div 6\}]$$

3. **Step 3: Evaluate operations inside curly brackets using BODMAS (Division before Addition).** First, perform the division:

$$30 \div 6 = 5$$

Substituting this:

$$10 - [8 - \{11 + 5\}]$$

Next, perform the addition:

$$11 + 5 = 16$$

Substituting this back:

$$10 - [8 - 16]$$

4. **Step 4: Simplify inside the square brackets.**

$$8 - 16 = -8$$

Substituting this back:

$$10 - [-8]$$

5. **Step 5: Perform the final subtraction.**

$$10 - (-8) = 10 + 8 = 18$$

Answer: 18

Solutions compiled for the notebook "Comprehensive Fundamentals of Integer Mathematics"